

Code:

```
import java.util.Scanner;

class Process {
    private String name;
    private int clock;
    private Scanner scanner;
    private Process otherProcess;
    private static final Object inputLock = new Object();

    public Process(String name, int initialTime) {
        this.name = name;
        this.clock = initialTime;
        this.scanner = new Scanner(System.in);
    }

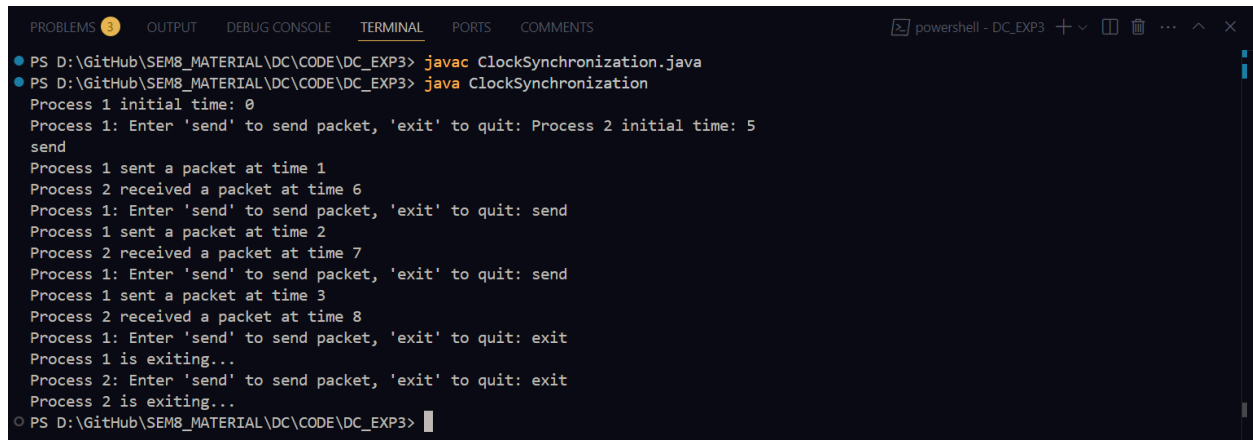
    public void setOtherProcess(Process otherProcess) {
        this.otherProcess = otherProcess;
    }

    public void sendPacket() {
        clock++;
        System.out.println(name + " sent a packet at time " + clock);
        otherProcess.receivePacket(clock);
    }

    public void receivePacket(int senderClock) {
        clock = Math.max(clock + 1, senderClock + 1);
        System.out.println(name + " received a packet at time " + clock);
    }

    public void startProcess() {
        System.out.println(name + " initial time: " + clock);

        while (true) {
            synchronized (inputLock) {
                System.out.print(name + ": Enter 'send' to send packet, 'exit' to quit: ");
                String input = scanner.nextLine().trim().toLowerCase();
            }
        }
    }
}
```


Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS powershell - DC_EXP3
PS D:\GitHub\SEM8_MATERIAL\DC\CODE\DC_EXP3> javac ClockSynchronization.java
PS D:\GitHub\SEM8_MATERIAL\DC\CODE\DC_EXP3> java ClockSynchronization
Process 1 initial time: 0
Process 1: Enter 'send' to send packet, 'exit' to quit: Process 2 initial time: 5
send
Process 1 sent a packet at time 1
Process 2 received a packet at time 6
Process 1: Enter 'send' to send packet, 'exit' to quit: send
Process 1 sent a packet at time 2
Process 2 received a packet at time 7
Process 1: Enter 'send' to send packet, 'exit' to quit: send
Process 1 sent a packet at time 3
Process 2 received a packet at time 8
Process 1: Enter 'send' to send packet, 'exit' to quit: exit
Process 1 is exiting...
Process 2: Enter 'send' to send packet, 'exit' to quit: exit
Process 2 is exiting...
PS D:\GitHub\SEM8_MATERIAL\DC\CODE\DC_EXP3>
```